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**Design of an Alumni Database System for Outcome-Based Evaluation and Institutional
Networking in Civil, Computer, and Electrical Engineering**

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February 2026

Chapter 1

In contemporary higher education, institutions are expected to demonstrate the effectiveness of their academic programs through measurable graduate outcomes. This mandate aligns with Commission on Higher Education (CHED) Memorandum Order No. 46, Series of 2012, which institutionalized Outcomes-Based Education (OBE) across Philippine higher education institutions (CHED, 2012). Under this framework, universities must articulate intended learning outcomes and provide evidence that these outcomes are achieved beyond graduation.

Global perspectives further highlight the importance of graduate tracking systems. The World Bank (2022) identifies employability as a key indicator of higher education quality, particularly in developing economies, while the Asian Development Bank (2021) emphasizes digital data systems as essential tools for strengthening governance and evidence-based decision-making. Moreover, accreditation frameworks aligned with the Washington Accord stress documented graduate attributes and continuous improvement processes (International Engineering Alliance [IEA], 2021). As the Philippine Technological Council (PTC), through ACBET, adopts these standards, institutions must present verifiable evidence of graduate achievement and professional readiness (Philippine Technological Council, 2023).

For the College of Engineering at Southern Luzon State University (SLSU), which aims to pursue PTC–ACBET accreditation by 2027, the development of a structured and optimized alumni database system is therefore a strategic priority. Such a system will enable systematic tracking of employment outcomes, support accreditation requirements, and strengthen outcomes-based evaluation and institutional planning.

Background and Definition of the Project

The pursuit of accreditation from the Philippine Technological Council (PTC) and the Accreditation and Certification Board for Engineering and Technology (ACBET) represents a strategic imperative for engineering institutions in the Philippines seeking to demonstrate educational quality and global competitiveness. As Southern Luzon State University (SLSU) aims to achieve PTC-ACBET accreditation by 2027, the institution must demonstrate compliance with rigorous outcome-based education (OBE) standards that require systematic evidence of graduate performance and professional development. According to the PTC, accreditation serves as a mechanism to ensure that engineering programs produce graduates who meet both national professional standards and international benchmarks, thereby enhancing the global competitiveness and employability of engineering graduates (Philippine Technological Council, 2023). This accreditation framework, which aligns with the Washington Accord to which ACBET holds provisional signatory status, necessitates that institutions implement robust data collection systems to track and evaluate alumni outcomes as evidence of program effectiveness (International Engineering Alliance, 2023). The ability to systematically demonstrate graduate success through empirical data has become not merely desirable but essential for meeting the documentation requirements mandated by accrediting bodies.

Despite these accreditation requirements, the College of Engineering at SLSU does not yet have a dedicated and centralized system for tracking alumni outcomes over time. At present, alumni data collection is done through Google Forms and managed by the Office of Student Affairs and Services (OSAS). While this method allows the University to gather some information, it remains limited and difficult for the College to fully utilize. Many records are

incomplete, response rates are relatively low, and contact details of graduates are often outdated, making follow-ups challenging. In addition, the College only receives the number of respondents rather than access to the complete dataset, which restricts deeper analysis. Because of these limitations, the College finds it difficult to thoroughly assess employment patterns, professional certifications, licensure performance, and the overall career progression of its graduates.

Southern Luzon State University (SLSU) currently facilitates an outcome-based evaluation of its alumni, focusing on graduates from 2018 to track their professional development and employment outcomes. At present, data collection is conducted through Google Forms and is managed by the Office of Student Affairs and Services (OSAS) through a spreadsheet. The data collected only sits on the spreadsheet as they do not have any centralized system that compiles and analyzes the records. At present, the SLSU Office of Student Affairs and Services (OSAS) maintains alumni records only for graduates from 2019 to 2022. The records show that from 2019 to 2022, there were only 48 respondents from civil engineering, 59 from computer engineering, and 80 from electrical engineering, which implies a great discrepancy from the total number of graduates.

In addition to OSAS, SLSU has an external alumni office, the **SLSU Alumni Federation, Inc.**, which maintains records of the university alumni using traditional spreadsheets. While the office collects basic member information and payment collections, it does not implement formal programs for alumni engagement, nor does it systematically track outcome-based evaluations or career achievements as it is not obliged to do so. Records from the SLSU Alumni Federation Inc. (AFI) show a significantly higher number of alumni responses, as they have been collecting data from 2018 up to the present. The total number of respondents

includes 386 from Civil Engineering, 289 from Computer Engineering, and 486 from Electrical Engineering. However, alumni typically accomplish the Google Form within their year of graduation as a requirement for obtaining their alumni ID, which explains why responses are concentrated at the point of graduation rather than updated over time. As a result, the current status of alumni remains largely unknown, leaving a significant gap in data critical for program assessment and institutional improvement.

Moreover, the **SLSU Engineering Alumni League (SEAL)** provides updates and announcements regarding engineering graduates through social media platforms, such as Facebook. These updates include accomplishments like board exam results, job hunting, and contributions to the university. However, SEAL lacks a centralized database or systematic records to monitor alumni achievements and career progression, which limits the College of Engineering's ability to evaluate the long-term success of its graduates.

At present, the College of Engineering does not have direct access to alumni data gathered through online surveys facilitated by Office of Student Affairs and Services (OSAS). This lack of access and systematic tracking capability directly restricts the College's ability to conduct the outcome-based evaluation that is fundamental to ACBET accreditation requirements. The ACBET accreditation criteria explicitly mandate that engineering programs demonstrate evidence of graduate achievement through systematic assessment of educational outcomes, requiring institutions to implement continuous improvement mechanisms based on reliable data collection and analysis (Accreditation and Certification Board for Engineering and Technology, 2023). Without a dedicated system for alumni tracking and outcome evaluation, the College cannot adequately document program effectiveness or demonstrate compliance with these

accreditation standards. Recognizing these limitations, the researchers aim to develop a dedicated alumni database system.

This capstone project addresses the existing gap in alumni tracking through the development of an alumni database system that is cost-effective, systematic, and optimized for decision-making support. The proposed system is designed to operate primarily as an offline database hosted within the College, while also allowing the synchronization of alumni data collected online via spreadsheets generated from survey tools such as Google Forms. Offline systems so far have worked based on several techniques. Among the most advanced techniques is MS Synchronize Framework or shortly MS Sync Framework, which is a platform from Microsoft for synchronizing data from many units and offline Web applications. (Cico, 2012).

In terms of data collection, Google Forms will serve as the primary tool because it is convenient, accessible, and cost-effective. Utilizing Google Forms eliminates the need to immediately develop a fully online system that would require paid hosting and continuous subscription costs. This practical approach allows the College to gather alumni data efficiently without incurring additional financial burden. The proposed system will also include a built-in SMTP email messaging feature to directly contact alumni through their registered email addresses, enabling follow-ups and data verification; although this feature is limited to sending up to 500 emails per day, it remains sufficient for organized and scheduled communication.

Regarding sustainability, Google Forms will be used for as long as the platform remains available and functional. If it becomes unavailable in the future, the College may then consider transitioning to a fully online system that would allow alumni to directly update and manage

their records. Through this approach, the College maximizes existing resources while maintaining flexibility for future system development and expansion.

The proposed alumni database system adopts an offline database design primarily because of its cost-effectiveness. Having an offline database system can be more cost effective as there are no ongoing fees for cloud services. Along with no costs for continuous internet connection or bandwidth usage. Implementing an offline database system has several benefits, including high security, performance, and data control. Since the system functions independently of internet connectivity, it reduces the security risks associated with transmitting sensitive information online while also removing network latency, resulting in faster data access and response times. In addition, an offline database provides complete administrative control over data storage, access, and security measures, making it an efficient and cost-effective approach for organizations that value data security and operational independence (Taborowski, 2025).

This hybrid approach ensures affordability without compromising flexibility or scalability. The system will enable the College to centrally track the employment outcomes, career progression, and professional achievements of Civil, Computer, and Electrical Engineering graduates. By consolidating and organizing alumni information, it will provide actionable insights for curriculum evaluation, program enhancement, and institutional planning, while fostering stronger alumni engagement and professional networking opportunities.

The Design of an Alumni Database System for Outcome-Based Evaluation and Institutional Networking is envisioned as a centralized, secure, and decision-support-oriented platform for the alumni of Civil, Computer, and Electrical Engineering at Southern Luzon State University. According to Puerzer and Rooney (2002), a centralized "assessment tool" allows

engineering programs to collect longitudinal data that is otherwise lost in departmental silos. By consolidating data from Civil, Computer, and Electrical Engineering into one platform, the University can perform "Decision-Support" functions, such as identifying if certain curriculum areas (like Power Systems or Structural Analysis) require updates based on the collective professional success of graduates. This centralization is what transforms simple data entry into a decision-support-oriented tool, enabling the College Dean and chairpersons to identify employment trends and skill gaps across all engineering disciplines simultaneously. The system will collect, store, manage, and analyze alumni data while maintaining restricted access for authorized personnel only, including the College Dean, college chairpersons, and system administrators. Individual alumni will not have personal accounts or direct access to the system, ensuring data privacy while enabling the College to maintain comprehensive oversight of graduate performance and achievements.

Beyond the creation of a database, this project focuses on the design and optimization of a socio-technical system that integrates people, processes, and data to support outcome-based evaluation and institutional networking. By applying methodologies, including process mapping, needs assessment, data flow diagramming, and prototyping, the study aims to provide the College with a sustainable and analytically robust tool for evaluating alumni performance and demonstrating program effectiveness to accrediting bodies. The system is expected to enhance institutional decision-making and contribute to the local, national, and global competitiveness of SLSU's engineering programs by providing the systematic evidence of graduate success that is essential for achieving and maintaining PTC-ACBET accreditation by 2027.

Statement of Project Objectives

To design, develop, and optimize a centralized alumni database system for the Civil, Computer, and Electrical Engineering programs at Southern Luzon State University, and to evaluate its effectiveness in supporting outcome-based evaluation and institutional networking.

Specific Objectives

1. To determine the features integrated in a good university alumni database.
 - 1.1 Data Insights Dashboard
 - 1.1.1 Reports and Exports
 - 1.2 Alumni Directory
 - 1.2.1 Search Engine
 - 1.2.2 Full CRUD (Create, Read, Update and Delete) Operation
 - 1.3 Segmentation and Filtering
 - 1.3.1 By Year Graduated Categorization
 - 1.3.2 By Program Categorization
 - 1.3.3 By Field of Specialization/Expertise
 - 1.3.4 By Area/Location of Current Job
 - 1.4 Cloud Synchronization
 - 1.5 Built-in Messaging System
 - 1.5.1 Integration with GMail
 - 1.6 Scalability and Performance

2. To design and develop a functional offline alumni database system that integrates multiple data sources into a centralized repository.

2.1 Number of Data Fields Integrated

2.1.1 Alumni Personal Information

2.1.2 Academic Background

2.1.3 Program Relevance and Curriculum Alignment

2.1.4 Research and Project Outputs

2.1.5 Micro-Credentials

2.1.6 Professional Credentials

2.1.7 Career Progression

2.1.8 Awards and Recognitions

2.1.9 Community and Industry Engagement

2.2 Number of Data Sources Consolidated

2.2.1 Responses from Google Forms

2.2.2 Recorded Data in the Spreadsheet

2.2.3 List of Graduates from the Dean's Office

2.3 Data Integrity

3. To evaluate the usability and acceptability of the developed alumni database system as perceived by end-users.

3.1 System Usability Scale (SUS)

3.1.1 Ability to compute OBE metrics

- 3.1.1.1 Percentage of Board passers
 - 3.1.1.2 Percentage of Employed alumni
 - 3.1.1.3 Percentage of In field-related jobs
 - 3.1.1.4 Percentage of In supervisory/managerial roles
 - 3.1.2 Accessibility of alumni performance data
 - 3.1.3 Support for program outcome measurement
- 3.2 User Satisfaction Rating
- 3.3 Ease of Data Entry and Retrieval

Scope and Project Constraints

This study focuses only on the design, optimization, and implementation of an offline alumni database system for the College of Engineering at Southern Luzon State University. It is specifically designed to track and analyze the career outcomes and professional milestones of graduates from the Civil, Computer, and Electrical Engineering programs. The research prioritizes the data of graduates from the year 2018 to facilitate a five-year post-graduation evaluation, which is considered a stable reflection of program effectiveness. By applying Industrial Engineering methodologies such as process mapping and data flow diagramming, the project aims to integrate people, processes, and data into a sustainable socio-technical system that supports institutional decision-making and curriculum enhancement.

In addition, this study focuses on consolidating fragmented data sources, including records from the Office of Student Affairs (OSAS), the SLSU Alumni Federation, Inc., and the SLSU Engineering Alumni League (SEAL). While these offices maintain various levels of

alumni information, this research concentrates on creating a centralized repository that the College of Engineering can independently access for academic and accreditation purposes. The system is intended to operate as a local administrative tool rather than a public social platform, ensuring that data management remains under the direct oversight of the College Dean and department chairpersons.

The constraints of the study involve limited user access, as individual alumni will not have personal accounts or direct access to the system, which ensures data privacy but limits real-time personal updates from the graduates themselves. While existing institutional surveys and social media announcements are utilized, they serve only as a foundational guide for the research. To ensure direct data access and accuracy, the study implements a custom-designed Google Form as the primary data collection tool. To streamline the process, respondents are only required to complete this single form. Utilizing conditional branching logic, the form automatically presents a specific set of questions based on the initial option selected by the alumni, ensuring that each participant only interacts with content relevant to their specific status or records. Furthermore, although the system is an offline solution for the College, it does not require manual data importation for maintenance; instead, it is designed to synchronize automatically with the primary data source to ensure the records remain current and efficient with minimal administrative intervention.

While the system handles data efficiently, it does not provide real-time updates for alumni access, as it is specifically designed as a specialized tool for internal accreditation purposes. To maintain this controlled environment, the synchronization feature is designed for simplicity, requiring only one click and offering three options: (1) pull data from the spreadsheet

into the system, (2) push data from the system into the spreadsheet, or (3) fully synchronize both sources to ensure consistency. Internet access is required only during the synchronization process; otherwise, the system functions entirely offline. This setup allows the College to maintain operational independence while still benefiting from flexible and efficient data integration when needed. To support long-term data integrity and scalability, cloud-based storage is provided through the university's Management Information Systems (MIS) office, with the understanding that any associated service fees or subscription costs are to be shouldered by the College of Engineering administration as needed.

Significance of the Project

The significance of the project extends to various stakeholders, including the Chairpersons of Civil (CE), Computer (CPE), and Electrical Engineering (EE), the Dean of College of Engineering (CEN), SLSU Engineering Alumni League (SEAL), Alumni, and future researchers.

For the **Chairpersons of CE, CPE, and EE**, the proposed alumni tracking database system will provide a centralized and reliable source of alumni information to support outcome-based evaluation, curriculum assessment, and continuous program improvement. The system will enable systematic alumni tracking of graduate employability, career progression, and professional achievements, which are significant in aligning academic programs with industry needs and accreditation requirements.

For the **Dean of CEN**, the project will serve as a decision-support tool that enhances academic planning and accreditation readiness. By consolidating alumni data into a structured

and accessible database, the system will support data-driven decision-making, strengthen monitoring of program performance, and facilitate the preparation of evidence required for outcomes-based education and engineering accreditation.

For the **SLSU Engineering Alumni League (SEAL)**, the system will support alumni networking and coordination by providing structured and updated alumni information. This will help in promoting professional collaboration, job opportunity dissemination, and stronger institutional linkages among engineering graduates.

For the **Alumni**, the system will provide a structured means for documenting professional achievements and career milestones. The study may also support networking initiatives such as job postings and professional announcements shared within the alumni community through SEAL-facilitated activities, strengthening alumni engagement and visibility.

For **future researchers**, this project may serve as a reference for studies related to Industrial Engineering-driven information systems, alumni tracking, and outcome-based evaluation in higher education. The methodologies, system design approach, and findings of this study may provide a foundation for further enhancements and adaptations of similar systems in other academic institutions.

Definition of Terms

The following terms used in this study will be defined both operationally and conceptually.

Accreditation Readiness - refers to an institution's ability to maintain full, ongoing compliance with accreditor standards, ensuring that documentation, outcomes, and curriculum alignment are always audit-ready.

Career Progression - The process of making progress in your career, for example by getting a better job or being promoted.

Centralized Repository - a secure, unified system that stores all critical regulatory documents, policies, and related data in one accessible location.

Cloud-Based Storage - is a mode of computer data storage in which digital data is stored on servers in off-site locations.

Cloud Synchronization - refers to the synchronization between devices for files that live on cloud storage.

Curriculum Assessment - is a process of gathering and analyzing information from multiple sources in order to improve student learning in sustainable ways.

Database - An organized collection of information set up for easy access, management, and updating, typically storing aggregations of data records.

Graduate Employability - refers to a graduate's ability to secure and maintain employment, which involves possessing a mix of skills, knowledge, and attributes that make them suitable for the job market.

Institutional Networking - A complex concept and abstract that we can relate to the way in which different social constructions interact with each other for common purposes.

Offline Database System - A database system that is stored locally on a device or server and does not require continuous access to the Internet

Outcome-Based Education (OBE) - A learner-centered approach to education that focuses on what a student should be able to do in the real world upon completion of their course or program.

Socio-Technical System - in organizational development is an approach to complex organizational work design that recognizes the interaction between people and technology in workplaces.

System Optimization - Involves making an existing system more efficient and effective without fundamentally altering its core structure or functionality.

Chapter II

REVIEW OF LITERATURE

Related Literature and Studies

The following review of related literature and studies is organized according to the sequence of the study's specific objectives to establish a coherent theoretical and empirical foundation for the design, development, and evaluation of the alumni database system.

I. Alumni Data Management in Higher Education Institutions

The first objective of this study requires a thorough understanding of how alumni data is currently managed in higher education institutions, including the methods of collection, storage, accessibility, and workflow efficiency. The literature below examines the nature of alumni databases, the challenges institutions face in maintaining them, and the consequences of fragmented data systems.

According to Sharada (2025), an alumni database for schools and universities must collect the right amount of the right information. Every detail is not needed, just the information that will strengthen connections and deliver value. That being said, the following are the list of some types of information that is generally prioritized: basic Contact Information, Education Details, Career Information, Preferred Communication Channels. This aligns with Chowdhury (2025) who stated that an alumni database is a centralized system that stores detailed information about your graduates, former members, or participants. This includes names, contact details, graduation year, current job information, and engagement activity. This implies that having a

well-organized alumni database is crucial. It brings all alumni details and interaction histories into one place, creating detailed profiles that help build stronger connections. scale.jobs (2025)

Furthermore, Chowdhury, T. (2025) stated that an alumni database is a centralized system that stores detailed information about graduates or former members, including names, contact details, graduation year, current employment information, donation history, and engagement activities. The consolidation of alumni information in a central platform enables institutions to manage alumni relations more effectively and support long-term engagement strategies.

While the ideal state of alumni data management involves centralized and structured systems, the reality for many institutions remains fragmented and inefficient. Research by Dai and Lan (2021) through Atlantis Press identifies that "fragmented alumni information" is a major barrier to institutional development. This fragmentation results from the use of disconnected tools — such as spreadsheets, social media, and manual records — which prevent institutions from obtaining a comprehensive and accurate picture of their graduates' outcomes. According to Johansyah et al. (2025), a robust alumni database is a fundamental pillar of graduate quality management, as it serves as a primary source for collecting the empirical data necessary to evaluate higher education programs. The effectiveness of alumni data management is critical for the continuous improvement of educational standards, ensuring that institutions can maintain an accurate and longitudinal record of their graduates' professional achievements.

As identified by Pasaribu et al. (2020), tracking alumni through a systematic database provides a "bridge" that connects the academic institution to the evolving requirements of work,

business, and industry. This data-driven approach allows for a sophisticated analysis of the association between a student's field of study and their eventual job description, which is essential for determining the long-term relevance of a university's curriculum. Cordova et al. (2021) emphasized that the Alumni Information System contributes a huge help to institutions and alumni that will allow them to reach out with each other, facilitating ongoing activities and accurate communication.

Without effective data management, universities face significant challenges in institutional networking and longitudinal tracking. As noted by Eslit (2023), maintaining an updated and centralized record is the only way to transform an "uncharted odyssey" of alumni into a clear roadmap for future institutional policy and program development. The challenge of alumni data fragmentation is not unique to developing nations. Hooge, E. (2020) observed that even well-resourced European universities struggle with incomplete alumni records due to inconsistent data collection at the point of graduation and a lack of institutional policies mandating alumni tracking. They recommend that institutions conduct a formal data management maturity assessment before designing any alumni information system, to identify specific gaps in data collection methods, storage infrastructure, data completeness, and workflow efficiency.

II. Features and Design Considerations of University Alumni Database Systems

The second objective seeks to establish a benchmark for what constitutes a well-designed alumni database system. The following literature examines the essential features, functionalities, and design considerations that enable effective alumni data management. Hasaranga et al .

(2024) emphasized that the integration of data visualization dashboards and data analytics in educational information systems significantly improves decision-making efficiency, as administrators can generate real-time reports, monitor key performance indicators, and export data for accreditation documentation — functionalities that correspond directly to the Data Insights Dashboard and Reports/Exports.

According to Agarwal (2023), to manage alumni data effectively, Database Management System (DBMS) or CRM is needed with a set of features that can handle the complexity of alumni data and engagement. Key features are: custom data fields, data validation and verification, alumni directory, communication tools, segmentation and filtering, integration with external platforms, reporting and analytics, data import and export, scalability and performance, and cloud-based options. Sahara (2025) agreed to this as she also mentioned that there is a need for a platform that is scalable, secure, and user-friendly, while meeting the needs of your institution. At this juncture, you'll want to consider features like data import/export, reporting capabilities, communication tools, and integration with other systems. In terms of communication, Scale.jobs (2025) stated that a strong messaging system is key to keeping your alumni network connected and engaged.

According to Etcuban and Durano (2015), the study aimed to analyze, design, and develop an alumni database to address the challenges encountered by faculty members and administrators in tracing graduates. Specifically, the objectives were to: (1) identify the problems encountered by faculty members and administrators when tracing the status of graduates; (2)

determine the features of a good university alumni database; (3) establish its design and development; and (4) evaluate the system users' assessment.

On Cloud Synchronization and Scalability: Kavis, M. J. (2014) argues that cloud-based architectures enable educational institutions to scale their data storage and processing capabilities without significant infrastructure investment, while ensuring data availability across multiple access points — which supports the Cloud Synchronization and Scalability features.

III. System Content: Alumni Data Fields for Outcome-Based Evaluation

The third objective focuses on the actual design and development of the alumni database system, including the integration of multiple data fields and data sources. The literature below provides the empirical basis for the content that the system must capture — particularly the data fields related to program relevance, professional achievement, and career progression — which together enable outcome-based evaluation.

A. Program Relevance and Curriculum Alignment

Program relevance serves as a primary determinant of graduate employability and career sustainability. The alignment between academic preparation and industry requirements reflects the effectiveness of curriculum delivery.

The perceived relevance of the BS Computer Engineering program to graduates' current jobs was generally rated as high, with a large proportion of respondents indicating that their

education was much to very much aligned with their employment requirements (Dita, 2024). However, a smaller group of graduates perceived limited applicability of their academic training, which may be attributed to differences in job roles or industry demands. Similar findings were reported by Lopez and Garcia (2019), who noted that while engineering graduates often recognize the strong theoretical foundation provided by their programs, some experience a disconnect between academic preparation and the evolving needs of the industry.

According to a tracer study conducted by Carigma et al. (2020) at the University of Rizal System, school-related factors—specifically curriculum and faculty instruction—emerged as the most significant contributors to the employability of Bachelor of Science in Computer Engineering graduates.

As emphasized by Salabi et al. (2024), the success of a curriculum is measured by the employability and professional competencies of its alumni, which serves as a basis for assessing the alignment between academic training and industry expectations. For instance, analyzing the career pathways and entry-level experiences of alumni helps institutions pinpoint specific knowledge gaps or "mismatches" between academic training and industry expectations. Consequently, tracer study data acts as a feedback mechanism that informs the revision of course content and the overall optimization of curriculum delivery to ensure that graduates remain competitive. Ultimately, as identified by Pasaribu et al. (2020), these studies transform alumni feedback into strategic insights, enabling universities to proactively adapt their programs to meet both local and global industry standards.

B. Professional Achievement and Career Progression

Professional achievement reflects the long-term effectiveness of educational preparation and is often measured through employment status, licensure performance, and upward mobility. These data fields — professional credentials, career progression, and employment outcomes — constitute the core alumni information that the database system must capture and organize to fulfill its OBE function.

According to a study conducted by Abana et al. (2019) at the University of Saint Louis, graduates in civil, electrical, and electronics engineering typically secure employment after six months due to licensure examination reviews, whereas computer engineering graduates tend to find work more quickly because they lack a board exam requirement and often prioritize industry certifications. The University Engineering graduates demonstrate high employability, typically securing positions shortly after passing their licensure examinations for the Civil and Electrical programs or immediately following graduation for the Computer Engineering program. This success is attributed to the university's holistic training approach, which effectively equips students with a combination of soft and hard skills specifically tailored to meet current industry demands. Consequently, the institution has shown consistent progress in producing competent professionals who are prepared to function effectively within their respective engineering sectors.

Electrical and Computer Engineering trajectories have been notably influenced by the expansion of data centers and IoT. Career progression in the 2020s has shifted toward a "hybrid" model where technical tenure is supplemented by specialized certifications. Multidisciplinary

Digital Publishing Institute (2021) identifies two forms of career success: objective (salary and hierarchical level) and subjective (job satisfaction and career adaptability). For engineering alumni, the transition from staff roles to supervisory or managerial positions often occurs within 5–8 years of graduation, facilitated by advanced professional certifications and lifelong learning initiatives (RSIS International, 2025).

According to Sicuan et al. (2023), professional achievement among graduates is often benchmarked by their performance in licensure examinations and their subsequent entry into the workforce, particularly in regulated professions like teaching. Their study highlights a consistent trend where higher academic performance, reflected in General Weighted Averages (GWA), moderately correlates with better licensure ratings, which in turn acts as a gateway to stable career paths. Furthermore, career progression is defined by the trajectory of graduates from entry-level positions to roles with greater responsibility and better compensation. Dassun et al. (2025) states that, longitudinal tracking of graduates allows institutions to assess not only immediate employment but also the long-term career growth of alumni, such as promotions and professional stability within their chosen fields.

The attainment of professional success is also heavily influenced by the continuous acquisition of specialized skills and advanced education. According to Patulin et al. (2024), graduates from advanced programs often demonstrate enhanced competitiveness, which facilitates their upward mobility in both domestic and global labor markets. It is also evident that graduates' career progression reflects both their personal and professional development, expanded employment opportunities, and deeper knowledge within their selected industries

(Oliva & Acledan, 2021). This is noted by Salabi et al. (2024), who state that the effectiveness of an educational program is validated when its alumni exhibit high-quality professional management and leadership skills in their workplaces. Additionally, the "uncharted odyssey" of career development involves a transition phase where graduates leverage their academic preparation to navigate professional challenges and achieve career milestones, emphasizing the role of higher education as a lifelong foundation for professional achievement.

C. Data Integrity

Ensuring the integrity of data within the alumni database system is essential for reliable outcome-based evaluation. According to Elmasri, R. and Navathe, S.B. (2016), data integrity in relational databases is enforced through domain constraints, key constraints, referential integrity, and semantic integrity rules embedded within the Entity-Relationship (ER) model. As the system must ensure that alumni records are accurate, consistent, and non-redundant across all integrated data sources.

IV. Outcome-Based Evaluation, System Effectiveness, and Institutional Networking

The fourth objective evaluates whether the developed system is usable, acceptable, and effective in supporting outcome-based evaluation. The literature below examines tracer studies as OBE tools (which underpin the metrics the system must compute), alumni database effectiveness (which defines what a successful system looks like), institutional networking (which is a key system output), and system usability evaluation (which provides the assessment methodology).

A. Tracer Studies as Outcome-Based Evaluation Tools

Tracer studies function as a structured feedback mechanism for evaluating program effectiveness and institutional accountability. The alumni database system must be capable of computing OBE-related metrics derived from tracer study data — including employability rates, job-field alignment, and career progression.

In the context of modern academic research, a tracer study serves as a vital tool for Outcome-Based Evaluation (OBE), acting as a feedback mechanism to assess the effectiveness and quality of higher education programs based on the real-world performance of their graduates. According to Johansyah et al. (2025), tracer studies are essential for graduate quality management, as they provide empirical data to evaluate whether the intended educational outcomes are being achieved post-graduation. According to a study conducted by Rohman et al. (2022), tracer studies serve as a vital tool for universities to enhance both formal and non-formal educational services, particularly in areas such as internship programs, problem-solving, and communication skills.

Tracer studies offer detailed and in-depth information on the job match, both horizontally (across fields of study) and vertically (across educational levels). Thus, tracer studies can help address issues of employment disparities and efforts for improvement. For universities, information on relevant competencies for the workforce can support curriculum and learning system enhancements (Adistana et al., 2024). According to Quitevis et al. (2019), tracer study results could help assess the usefulness of the programs as well as their effectiveness, and eventually its success or failure.

Tracer studies from 2024 and 2025 indicate that engineering graduates consistently maintain an employability rate exceeding 80% within the first year of graduation (Aribe & Rojo, 2025). The evaluation process seeks to determine the relevance of the study program to the labor market, identifying potential "mismatches" between academic training and industry requirements (Pasaribu et al., 2020; Patulin et al., 2024). Ultimately, this form of evaluation extends beyond mere employment rates to include holistic outcomes like life satisfaction and active citizenship (Tutor et al., 2019).

B. Alumni Database System Effectiveness

The effectiveness of the developed system is measured not only by its technical functionality but also by its ability to support institutional decision-making and data-driven program evaluation. The following literature establishes the criteria for evaluating alumni database effectiveness — a dimension directly assessed through the SUS indicators and perceived usefulness measures.

The effectiveness of tracking alumni depends heavily on the transition from manual ledgers to intelligent digital systems. Research by Dai and Lan (2021) through Atlantis Press identifies that "fragmented alumni information" is a major barrier to institutional development. According to Salabi et al. (2024), the strategic implementation of an alumni database significantly enhances educational quality by providing the evidence required for institutional stakeholders to align educational delivery with market needs.

C. Institutional Networking

Institutional networking is a critical system output that is evaluated as part of the system's overall effectiveness. The system's ability to support alumni coordination, mentorship, and professional collaboration — assessed through user perception is grounded in the following literature.

A centralized alumni record is no longer just a storage tool but a platform for "Social Capital" development. Centralized records allow for the use of AI-powered tools, such as the AlumniConnect platform, which utilizes searchable directories and intelligent chatbots to foster real-time mentorship and job referrals (IRJET, 2025). This networking infrastructure is essential for "Graduate Quality Management," as it centralizes feedback from various disciplines (Johansyah et al., 2025).

According to Sankaran and Rath (2021), the impact of alumni achievement is critical to an academic institution's reputation, as successful graduates serve as enduring role models for current students. Consequently, active alumni participation and their specific recommendations are indispensable elements for the successful assessment and enhancement of an engineering curriculum. Garača and Šimić (2016) emphasized that tracking alumni career paths provides empirical data that helps universities align their curriculum with the actual needs of the labor market.

D. System Usability Evaluation

A critical component of Objective 4 is the evaluation of the system's usability and acceptability using the System Usability Scale (SUS). The following literature provides the theoretical and methodological basis for this evaluation approach.

Bangor, Kortum, and Miller (2008) conducted a large-scale empirical study validating the SUS across over 2,300 surveys spanning ten years, establishing adjective-based interpretations: scores between 85–100 correspond to "Excellent," 73–85 to "Good," 52–73 to "OK," and below 52 to "Poor." This adjective scale provides a more intuitive interpretation for non-specialist evaluators such as university administrators, making it appropriate for the system evaluation in this study.

Lewis (2018) further validated the SUS as a reliable and valid instrument for measuring perceived usability, noting that it demonstrates high internal consistency (Cronbach's alpha typically exceeding 0.90) and that its two-factor structure — capturing both usability and learnability — is well-suited for evaluating information systems in educational contexts where end-users may have varying levels of technical proficiency.

Related Studies

Several studies have undertaken the design, development, or evaluation of alumni-related information systems, providing both methodological and practical precedents for this research.

Cordova et al. (2021) proposed a model and framework for an alumni tracking and information system. They emphasized that the Alumni Information System contributes a huge help to institutions and alumni that will allow them to reach out with each other. The system also facilitates active or ongoing activities and has very accurate communication. Once the system is implemented successfully, it would create a harmonious and meaningful relationship between the institution and the alumni.

Garača and Šimić (2016) emphasized that tracking alumni career paths provides empirical data that helps universities align their curriculum with the actual needs of the labor market. Their study on improving study programme quality through an alumni tracking information system provides a direct precedent for the system being developed in this research.

Alegado (2015) designed and developed an alumni database system to address the challenges encountered by faculty and administrators in tracing graduates.

Fathoni, A. N., Agung, A. I., Fransisca, Y., Khotimah, K., & Saputra, B. S. (2025) used tracer study methodology to map the career pathways of electrical engineering education graduates, demonstrating the type of longitudinal analysis that the proposed system's OBE analytics module must be capable of performing. Their work reinforces the need for a system that can compute employability rates, job-field alignment, and career progression metrics using structured alumni data.

Conceptual Framework

This conceptual framework presents the development of an Alumni Database System using an Input–Process–Output (IPO) model with feedback mechanism to support outcome-based evaluation and institutional networking in Civil, Computer, and Electrical Engineering Alumni of Southern Luzon State University.

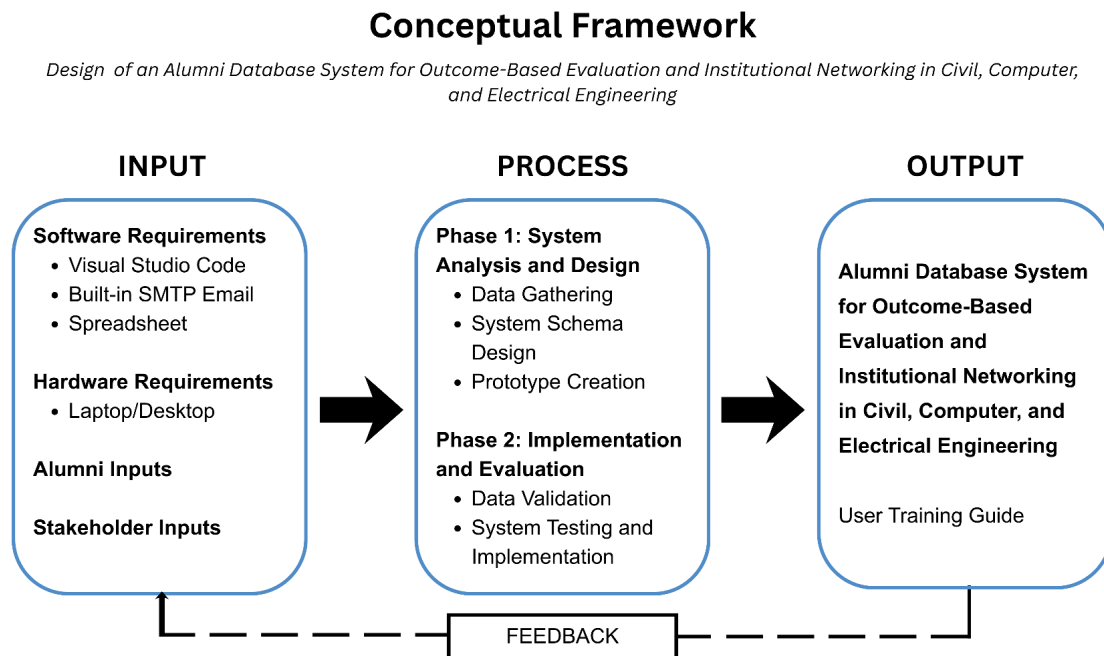


Figure 1. Conceptual Framework

The inputs include software and hardware requirements, as well as alumni and stakeholder data that define system needs and content. The process consists of two phases: system analysis and design (needs assessment, concept development, and prototyping) and implementation and evaluation (data validation, testing, and deployment). The output is a functional alumni database system with a user training guide, while the feedback mechanism ensures continuous improvement and system refinement.

Chapter III

METHODOLOGY

This study adopts a systems-oriented and Industrial Engineering (IE)–based methodology to design, develop, and optimize an alumni database system for outcome-based evaluation and institutional networking in the College of Engineering at Southern Luzon State University, specifically Civil, Computer and Electrical Engineering alumni. The methodology is divided into two major phases: (1) Systems Analysis and Design and (2) System Implementation and Evaluation. Each phase explicitly applies IE tools to ensure efficiency, effectiveness, and decision-support capability.

Project Design

This study employs a developmental and descriptive research design grounded in Industrial Engineering (IE) systems analysis and design principles. The developmental aspect focuses on the design, optimization, and implementation of an alumni database system for outcome-based evaluation, while the descriptive component examines the existing alumni data management processes and information gaps within Southern Luzon State University.

The research is conducted in two phases. Phase 1: Systems Analysis and Design applies IE tools such as needs assessment, stakeholder analysis, process mapping, data flow diagramming, and entity–relationship modeling to analyze the system requirements and the current alumni data processes.

Phase 2: System Implementation and Evaluation involves the development of the offline alumni database system with spreadsheet synchronization functionality, followed by system testing, data validation, performance evaluation, and process optimization using IE principles such as lean process improvement and key performance indicators (KPIs). The outcome of the research is a functional, cost-effective, and decision-support-oriented alumni database system that supports outcome-based evaluation and institutional networking within the College of Engineering.

Project Setting

The study is conducted at Southern Luzon State University (SLSU), specifically within the College of Engineering, which offers Civil Engineering, Computer Engineering, and Electrical Engineering programs. The College serves as the primary setting and beneficiary of the proposed alumni database system. Key institutional stakeholders involved in the study include the Dean of the College of Engineering, college chairpersons, and designated system administrators, who will act as authorized users of the system.

Additional contextual inputs are derived from existing alumni data practices of the Office of Student Affairs and Services (OSAS), the SLSU Alumni Federation, Inc., and the SLSU Engineering Alumni League (SEAL). Alumni data utilized in the study pertain to engineering graduates, particularly those who graduated in 2018, representing the target group for five-year post-graduation outcome-based evaluation. The proposed system will be hosted locally within the College of Engineering and will not be directly accessible to alumni, ensuring administrative control, data privacy, and institutional ownership of alumni performance data.

Respondents

The respondents of this study include the Dean of Engineering, the Program Chairpersons of Civil, Computer, and Electrical Engineering, and the alumni of the College of Engineering at Southern Luzon State University. The institutional stakeholders were selected through purposive sampling due to their direct involvement in alumni tracking, curriculum evaluation, and program outcome monitoring, and they provide the system requirements and evaluation criteria for the proposed alumni database system. Additionally, the alumni serve as the primary data sources, as all career outcome and professional achievement data originate from them, supplying the necessary information for system development and validation.

Data Sources

The following procedures outline how the researchers obtained the data relevant to the study:

1. A formal letter of request was submitted to the Dean of Engineering of Southern Luzon State University to obtain approval for the conduct of the study and access to relevant alumni-related records.
2. Interviews and consultations were conducted with the Dean of the College of Engineering, Program Chairpersons and selected administrative personnel involved in alumni data handling. These interviews aimed to gather information regarding existing alumni data management practices, challenges encountered in outcome-based evaluation and system requirements.

3. The researchers collected secondary alumni data from Google Form survey results, records from Office of Student Affairs and Services (OSAS), SLSU Alumni Federation, Inc., and publicly available alumni achievement announcements from SEAL to identify information gaps and define database content.
4. A review of institutional reports, academic journals, books, online articles, Tracer Studies and related studies on alumni tracking, outcome-based education, and Industrial Engineering-driven information systems was conducted.
5. All data gathered from interviews, secondary records, system testing, and literature review will be organized, encoded, and analyzed using appropriate Industrial Engineering tools and descriptive statistical techniques to support system optimization and evaluation.

Data Analysis

The data gathered for this study will be analyzed using a combination of Industrial Engineering (IE) tools and descriptive statistical techniques to ensure the system is optimized for decision-support and institutional networking. The analysis begins with Process Mapping, which will be utilized to visualize the current workflows of alumni data handling within the Office of Student Affairs (OSAS), the SLSU Alumni Federation, Inc., and the SLSU Engineering Alumni League (SEAL). This allows the researchers to identify bottlenecks and data gaps in the existing manual spreadsheets and traditional record-keeping systems. Parallel to this, a Needs Assessment will be conducted through interviews and consultations with the Dean and Program Chairpersons to translate institutional requirements into specific system functionalities, such as the hybrid data entry approach and restricted administrative access.

The structural design of the system will be analyzed using Data Flow Diagrams (DFD) and Entity-Relationship (ER) Modeling. The DFD will map the movement of information from external survey imports (such as Google Forms) into the centralized offline repository , while the ER model will define the logical connections between graduates' profiles, employment outcomes, and professional achievements to ensure data integrity. evaluate the system's effectiveness, the researchers will apply Performance Metrics and Optimization principles. This involves using lean process improvement to streamline data importation and establishing Key Performance Indicators (KPIs) to measure the system's ability to track five-year post-graduation outcomes for the 2018 cohort.

Finally, descriptive statistics, specifically frequency distributions and percentages, will be used to organize and summarize these metrics. This data-driven approach will provide the College of Engineering with actionable insights for curriculum enhancement and the evidence required for PTC-ACBET accreditation.

Timeframe

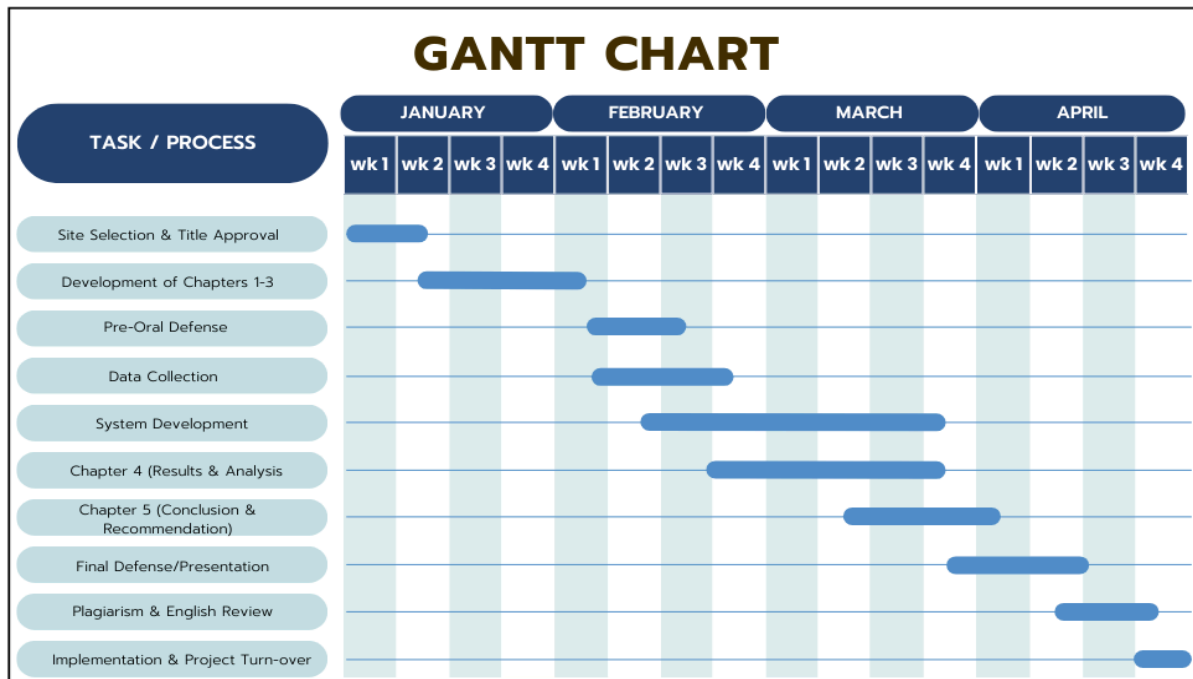


Figure 2. Project Gantt Chart

The researchers utilize the Gantt chart as a strategic management tool to monitor project development and ensure all tasks are completed by the established deadline. This visual overview provides a clear timeline for the Capstone Project, allowing the team to track progress across each critical phase. According to the schedule, the project is set to commence in January and conclude with the final implementation and turnover during the last week of April.

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